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AI-BASED PREDICTIVE AND VALUE-BASED STORAGE OPTIMIZATION
FOR MOBILE DEVICES

METHODOLOGY

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OPTIMIZATION FOR MOBILE DEVICES

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3.1 Introduction

This chapter presents the research methodology adopted for the design and development of the AI-Based Predictive and Value-Based Storage Optimization framework for mobile phones. Methodology in applied computing research provides the structured procedural framework through which a system is conceived, designed, implemented, and evaluated. A well-defined methodology ensures that research activities are systematic, reproducible, and aligned with the stated objectives.

The proposed system addresses a clearly identified gap: the absence of proactive, AI-driven storage management tools that combine time-series growth forecasting with value-based media ranking on mobile devices. The methodology has been selected to reflect both the research nature of the problem and the practical demands of system development within an academic environment at The Copperbelt University (CBU).

This chapter is organized as follows: Section 3.2 describes the overall research and development methodology. Section 3.3 covers information gathering and analysis activities, including literature review and iterative development sprints. Section 3.4 presents the requirements specification, comprising both user and system requirements. Section 3.5 provides a system analysis covering feasibility. Section 3.6 concludes the chapter.

3.2 Methodology

The study adopts a quantitative research approach embedded within an applied research framework. The quantitative orientation is appropriate because the performance of the proposed system will be evaluated using measurable, numerical metrics — specifically Mean Absolute Error (MAE), Root Mean Square Error (RMSE), forecasting accuracy, value-ranking precision, and storage lifespan extension ratios. These metrics lend themselves to objective comparison against a baseline heuristic approach.

For system development, the Agile software development methodology is adopted. Agile is characterized by iterative and incremental development cycles (sprints), continuous testing and integration, and responsiveness to feedback and refinement. It is well-suited to this project for several reasons:

- The system comprises multiple interacting components (data collection, AI prediction, value scoring, optimization, UI) that benefit from incremental integration rather than a single large-scale implementation.
- The AI models (ARIMA, regression-based forecasting, value scoring algorithms) require iterative refinement based on empirical performance data.
- The project timeline (February–July 2026) aligns naturally with short sprint cycles of one to three weeks.
- Agile allows early testing of individual modules before full system integration, reducing the risk of late-stage failures.

The research approach is applied in nature: it seeks to produce a functional prototype system that demonstrates measurable improvements over existing reactive storage. The research approach is applied in nature: it seeks to produce a functional prototype system that demonstrates measurable improvements over existing reactive storage management tools, rather than to generate purely theoretical contributions. The study is grounded in real-world usage data collected from consenting mobile device users at CBU.

The combination of quantitative evaluation and Agile development creates a methodology that is both scientifically rigorous and practically executable within the constraints of a final-year undergraduate project.

Dimension	Choice	Justification
Research type	Applied	Produces a functional prototype solving a real-world problem
Research approach	Quantitative	Performance measured using numerical metrics (MAE, RMSE, precision)

Dimension	Choice	Justification
Development methodology	Agile (Scrum-inspired)	Iterative sprints suit AI model refinement and modular development
Data collection	Primary (device logs)	Storage metadata and usage patterns collected from consenting CBU students
Evaluation strategy	Comparative	Proposed framework benchmarked against baseline heuristic approach
Deployment scope	Prototype	Experimental validation only; not commercial deployment

Table 1: Evaluation Table

3.3 Information Gathering and Analysis

Information gathering for this project spans two complementary activities: a structured literature review to establish theoretical grounding and identify the research gap, and iterative development sprints in which empirical data is collected and analyzed during system construction.

3.3.1 Literature Review

A systematic literature review was conducted to survey existing work on mobile storage management, predictive resource allocation, and value-based media classification. The review informed three aspects of the methodology:

- Identification of the research gap: Prior systems (Li et al., 2020; Chen et al., 2019; Park et al., 2018) rely on reactive, heuristic-based storage cleanup and do not integrate predictive forecasting with value-based ranking. This gap directly motivates the proposed framework.
- Selection of AI techniques: Time-series forecasting literature (Hyndman & Athanasopoulos, 2018; Zhang et al., 2019) supports the use of ARIMA and regression-based models for storage growth prediction. These models are computationally lightweight and appropriate for on-device execution.
- Design of the value scoring mechanism: Media classification literature (Torres et al., 2022; Szeliski, 2021) informs the selection of scoring features — recency (last accessed date), access frequency, and file size contribution — as proxies for media utility.

Sources consulted include IEEE Transactions on Mobile Computing, IEEE Access, Journal of Mobile Computing, and foundational machine learning texts (Goodfellow et al., 2016; Brownlee, 2017). The review was confined to publications from 2017 onwards to ensure relevance to contemporary mobile hardware and software environments.

Review Theme	Key Sources	Contribution to Design
Mobile storage management	Li et al. (2020); Chen et al. (2019)	Confirmed reactive nature of existing tools; validated research gap
Time-series forecasting	Hyndman & Athanasopoulos (2018); Zhang (2019)	Justified ARIMA/regression selection for storage growth prediction
Value-based media ranking	Torres et al. (2022); Kumar & Singh (2021)	Informed recency, frequency, size scoring feature selection
Mobile resource optimization	Rahman et al. (2020); Almeida et al. (2017)	Contextualized AI-driven proactive management approach

Review Theme	Key Sources	Contribution to Design
ML evaluation metrics	Goodfellow et al. (2016); Brownlee (2017)	Defined MAE, RMSE, precision, recall as evaluation framework

Table 2: Technology Review Table

3.3.2 Iterative Development Sprints

The development phase is structured into seven Agile sprints, each targeting a specific system component or integration milestone. Each sprint follows a cycle of planning, implementation, testing, and review before the next sprint commences. Primary data — storage usage logs, file metadata, and user feedback — is collected and analyzed within the relevant sprints.

Data collection procedure within sprints:

- Participants: Consenting mobile device users (Android smartphones) from The Copperbelt University student population. Informed consent forms will be administered prior to any data collection.
- Data collected: Storage usage logs (total storage, used storage, free storage over time), media file metadata (file size, file type, creation date, last accessed date, file path).
- Collection tool: A lightweight background data collection service implemented in Flutter, running on participants' devices during the data collection sprint.
- Duration: Storage logs collected over a minimum four-week observation period to capture sufficient temporal variation for time-series modelling.
- Privacy: All data is stored locally on each participant's device. No data is transmitted to external servers. Collected data is anonymized before analysis.

Sprint	Focus Area	Duration	Key Deliverables	Evaluation Criterion
1	Data Collection Module	2 weeks	Storage metadata harvester; SQLite schema; scan trigger	Data completeness; correct metadata extraction
2	Storage Prediction Engine	2 weeks	ARIMA/regression model; forecast output; MAE baseline	MAE and RMSE within acceptable thresholds
3	Value Scoring Engine	2 weeks	Scoring algorithm; weighted recency/frequency/size model	Precision of high-value vs low-value classification

Sprint	Focus Area	Duration	Key Deliverables	Evaluation Criterion
4	Optimization Engine	1 week	Ranked deletion plan generator; threshold logic	Accuracy of plan vs manual expert ranking
5	UI & Notification Module	2 weeks	Dashboard; recommendation screen; alert integration	Usability feedback from test users
6	Integration & Testing	3 weeks	End-to-end integrated system; unit + integration tests	System passes all functional test cases
7	Evaluation & Refinement	2 weeks	Baseline comparison; performance metrics report	Statistically significant improvement over baseline

Table 3: Data Collection Timeline

Each sprint concludes with a review against its defined evaluation criterion. Where performance targets are not met, the sprint is partially repeated or the subsequent sprint plan is adjusted — consistent with Agile principles of continuous improvement.

3.4 Requirements Specification

Requirement's specification translates the research objectives and stakeholder needs into precise, verifiable system requirements. Requirements are categorized into User Requirements (what the user needs to accomplish) and System Requirements (the technical capabilities the system must possess). Each requirement is assigned a unique identifier, priority level, and verification method.

3.4.1 User Requirements

User requirements describe the system from the perspective of the Mobile Device User — the primary stakeholder. These requirements were derived from the problem statement, research objectives, and the literature review findings regarding user behavior and storage management practices.

ID	User Requirement	Type	Priority	Verification
UR01	The user shall be able to view their current device storage usage at a glance on a dashboard	Functional	High	System test
UR02	The user shall receive a forecast of how many days remain before storage is full	Functional	High	System test
UR03	The user shall receive a ranked list of files recommended for deletion	Functional	High	System test
UR04	The user shall be able to approve or reject individual deletion recommendations	Functional	High	System test
UR05	The user shall receive a proactive notification when storage is forecast to reach a critical threshold	Functional	High	System test
UR06	The user shall be able to view the value score assigned to each file and understand why it was scored low	Functional	Medium	User acceptance test
UR07	The user shall be able to configure the storage threshold that triggers alerts	Functional	Medium	System test
UR08	The user shall retain full control — no file shall be deleted without explicit approval	Functional	High	System test

ID	User Requirement	Type	Priority	Verification
UR09	The user shall be able to view historical storage usage trends	Functional	Low	System test
UR10	The system shall not noticeably degrade device performance or battery life during operation	Non-Func.	High	Performance test

Table 4: User Requirement Table

3.4.2 System Requirements

Functional Requirements

ID	System Requirement	Type	Priority	Verification
FR01	The system shall collect storage metadata (file size, type, creation date, last accessed date) from the device file system	Functional	High	Unit test
FR02	The system shall store all collected metadata in a local SQLite database using the Room persistence library	Functional	High	Unit test
FR03	The system shall apply a time-series forecasting model (ARIMA or regression) to historical storage data to predict future storage growth	Functional	High	Unit test + MAE/RMSE evaluation
FR04	The system shall compute a composite value score for each media file based on recency, access frequency, and size contribution	Functional	High	Unit test + precision/recall evaluation
FR05	The system shall generate a ranked deletion plan combining forecast pressure and value scores	Functional	High	Integration test
FR06	The system shall trigger a device notification when the forecasted days-to-full falls below a configurable threshold	Functional	High	System test

ID	System Requirement	Type	Priority	Verification
FR07	The system shall present forecast data and recommendations through a dashboard UI built with Jetpack Compose	Functional	High	UI test
FR08	The system shall execute file deletions only upon receiving explicit user confirmation	Functional	High	System test
FR09	The system shall update the local database and re-run scoring after each approved deletion	Functional	Medium	Integration test
FR10	The system shall log model performance metrics (MAE, RMSE) for evaluation purposes	Functional	Medium	System test

Table 5: Functional Requirement Table

Non-Functional Requirements

ID	System Requirement	Type	Priority	Verification
NR01	The system shall operate entirely on-device with no internet connectivity required for core functions	Non-Functional	High	System test
NR02	The storage scan and scoring cycle shall complete within 30 seconds on a mid-range Android device	Non-Functional	High	Performance test
NR03	The system shall consume no more than 5% additional battery per day during background operation	Non-Functional	High	Performance test
NR04	All user data shall remain on-device; no personal data shall be transmitted externally	Non-Functional	High	Security review
NR05	The application shall be compatible with Android 10 (API level 29) and above	Non-Functional	High	Compatibility test

ID	System Requirement	Type	Priority	Verification
NR06	The UI shall conform to Material Design 3 guidelines with minimum 4.5:1 color contrast ratios	Non-Functional	Medium	UI audit
NR07	The system shall handle a file catalogue of up to 50,000 files without performance degradation	Non-Functional	Medium	Load test
NR08	The local database shall not exceed 50 MB of storage footprint for system data	Non-Functional	Low	System test

Table 6: Non-Functional Requirement Table

3.5 System Analysis

System analysis examines the feasibility of the proposed system across five dimensions: technical, economic, operational, schedule, and legal/ethical. Feasibility analysis validates that the proposed solution is achievable within the constraints of the project environment before significant development effort is committed.

Feasibility Area	Assessment	Justification
Technical	Feasible	Flutter, Python with Scikit-learn, and SQLite are mature, well-documented technologies available at no cost. ARIMA and regression models are computationally lightweight for on-device execution.
Economic	Feasible	All tools are open-source. Estimated project cost is K2,850 (data bundles, printing, contingency) as per the approved project proposal budget.
Operational	Feasible	Target users (CBU students) already use Android devices. The app requires no behavioral change beyond reviewing recommendations. Minimal training needed.
Schedule	Feasible	The 7-sprint Agile plan fits within the February–July 2026 project timeline defined in the Gantt chart.
Legal / Ethical	Feasible	Data collection requires informed consent. All storage data is processed locally on-device; no personal data is transmitted externally. Ethical clearance will be sought from CBU.

Table 7: Feasibility Study Table

Overall, the system is assessed as feasible across all five dimensions. The primary risk is the performance of the AI prediction model on devices with limited historical data (i.e., newly formatted devices). This is mitigated by falling back to a regression-based model with a shorter observation window when insufficient data is available for ARIMA modelling.

A secondary risk is user adoption: students may be reluctant to grant storage access permissions. This is mitigated by clear in-app explanations of data usage, local-only processing assurances, and the voluntary nature of participation.

3.6 Conclusion

This chapter has presented a comprehensive methodology for the design, development, and evaluation of the AI-Based Predictive and Value-Based Storage Optimization framework. The quantitative, applied research approach combined with Agile development methodology provides a rigorous yet flexible structure for delivering a functional prototype within the project timeline.

The information gathering activities — systematic literature review and iterative sprint-based data collection — ensure that the system is grounded in both theoretical evidence and real-world usage data from CBU mobile device users. The requirements specification provides 18 verifiable requirements (10 functional, 8 non-functional) that directly translate research objectives into implementable system features, with explicit priority levels and verification methods.

The feasibility analysis confirms that the proposed system is achievable using available open-source technologies (Flutter, Python, Scikit-learn, SQLite) within the approved project budget of K2,850 and the February–July 2026 development timeline. Identified risks — model cold-start on data-sparse devices and user permission reluctance — are addressed through defined mitigation strategies.

Together, these elements establish a solid methodological foundation from which Chapter 4 (System Design) proceeds to translate the requirements and analysis into a concrete system architecture, UML models, and physical design specification.

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